SMART INDIA HACKATHON 2025

Smart Traffic Management

- **Problem Statement ID – SIH25050**
- **Problem Statement Title - Smart Traffic Management system for urban congestion**
- **Theme- Transportation & Logistics**
- **PS Category- Software**
- **Team ID-**
- **Team Name - Pixel Engineers**

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Proposed Solution :

- **The Problem:** Major city intersections use outdated, fixed-timer traffic lights, leading to massive traffic jams, wasted fuel, and increased pollution.
- **Our Solution:** An AI-powered system that replaces these "dumb" timers with a dynamic, intelligent controller that sees the traffic and thinks in real-time.
  - **The Eyes (AI Perception):** We use existing city cameras and a **CNN model (YOLO)** to get a live, accurate count of vehicles and measure queue lengths in every lane.
  - **The Brain (AI Decision-Making):** A **Reinforcement Learning (RL) model** analyzes this data and makes smart, second-by-second decisions on signal timings to maximize traffic flow.

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TECHNICAL APPROACH

Technologies Used:

- **Software:** Python, OpenCV, PyTorch/TensorFlow, SUMO (for simulation).
- **Hardware:** Deployed on a central server or on edge computers at intersections.

How It Works:

- **Ingest:** Capture the live video feed from traffic cameras.
- **Analyze:** The CNN model detects and counts all vehicles, creating a data map of the intersection.
- **Decide:** The RL agent receives this data ("State") and chooses the optimal "Action" (e.g., extend the green light).
- **Act:** The system sends a direct command to the traffic light controller.

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FEASIBILITY AND VIABILITY

Feasibility:

- Highly feasible as it leverages **existing camera infrastructure**, reducing hardware costs.
- Uses proven, open-source AI models and simulation tools.

Challenges & Our Solutions:

- **Challenge:** Integrating with older traffic light hardware.
- **Solution:** Develop a modular API and simple hardware adapters.
- **Challenge:** Poor video quality in bad weather.
- **Solution:** Train the AI on augmented data (rain, fog, night) to ensure robustness

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IMPACT AND BENEFITS

Impact (How it helps people):

- **Reduces average commute time by over 10%**.
- Drastically cuts down on vehicle idling time and fuel consumption.
- Provides a city-wide, real-time **traffic dashboard for authorities**.

Benefits (The Big Picture):

- **Economic:** Saves citizens money on fuel and reduces productivity losses from gridlock.
- **Environmental:** Significantly lowers carbon emissions and air pollution.
- **Social & Future Potential:** The system **saves lives** by creating dynamic **"Green Corridors" for ambulances (Project Nightingale)**. It's a true **City Intelligence Platform**, enabling future uses like **predictive accident detection** and **data-driven urban planning**.

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RESEARCH AND REFERENCES

- **Simulation Tools:** SUMO (Simulation of Urban MObility).
- **Core Technologies:** OpenCV, PyTorch/TensorFlow, Reinforcement Learning (RL).
- **Field of Study:** AI for Smart Cities, Reinforcement Learning for Traffic Signal Control.

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